Current	Result	Time	Cycles	SM Frequency	Process	Attributes
Kernel: matmul_tiled	624 - matmul_tiled	2.78 ms	5,214,116	1.87 GHz	3 - NVIDIA GeForce RTX 4060 Laptop GPU	[NVLink:internal_clock,sm]

Summary | Details | Source | Compare | Comments | Session | Resources

Compare | Save | Close | Page | Export

GPU Speed Of Light Throughput

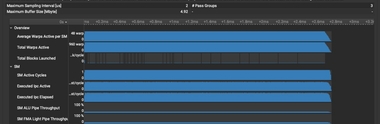
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for individual sub-metrics of Compute and Memory to clearly identify the higher utilized pipeline. Low throughput for both indicates that neither of the compute or memory resources of the GPU is utilized enough.

Compute [SM] Throughput [%]	96.70	Duration [ms]	2.78
Memory Throughput [%]	96.70	Cycles [cycle]	5214,116
L1/TEX Cache Throughput [%]	96.93	Ect Active Cycles [cycle]	5,232,888.50
L2 Cache Throughput [%]	96.57	Ect Frequency [Ghz]	1.87
DRAM Throughput [%]	1.38	DRAM Frequency [Ghz]	7.09

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory throughput. Refer to the Kernel Profiling Guide for further improvement opportunities, work will likely need to be shifted from the most utilized to another unit. Start by analyzing workloads in the Compute Workload section.

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 0% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Roofline topic for more details on roofline analysis.

GPU Throughput

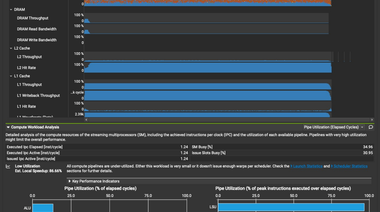


PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	2
Sampling Interval Avg Ratio [%]	4.62

Number Of Passes	3
Buffer Full	0



Compute Workload Analysis

Pipe Utilization (Shared Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.24
Executed Ipc Active [inst/cycle]	1.24
Issued Ipc Active [inst/cycle]	1.24

Low Utilization All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Instructions Retired [inst] 1,294,464 Block Limit SM 32
Block Limit Registers 32 Block Limit Shared Mem 32
Block Limit Warps 32 Theoretical Active Warps per SM 32
Theoretical Occupancy [%] 100 Achieved Occupancy [%] 96.70
Achieved Active Warps Per SM [warp] 30.94

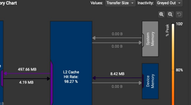
Section: Low Utilization

Warp Cycles Per Issued Instruction [cycle] 15.97
Warp Cycles Per Executed Instruction [cycle] 15.97
Avg. Active Threads Per Warp 32
Avg. Not Predicated Off Threads Per Warp 31.02

Section: Compute Workload Analysis

Pipe Utilization (% of elapsed cycles)

Pipe Utilization (% of peak instructions executed over elapsed cycles)



Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between the units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each unit are below.

Memory Throughput [Gbyte/s]	3.03	Mem Busy [%]	61.98
L1/TEX Hit Rate [%]	7.27	Max Bandwidth [%]	96.93
L2 Hit Rate [%]	96.57	Mem Pipes Busy [%]	96.70
Mem Hierarchy Bottleneck Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Size Saved [%]	0	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	0		

Memory Chart

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions from. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler issues one of them into the next instruction slot. During each cycle in which no eligible warp is ready, no instruction can be issued, leaving more cycles until a warp becomes eligible or new warps can be added to the pool.

Active Warps Per Scheduler [warp]	11.60
Eligible Warps Per Scheduler [warp]	1.64
Issued Warp Per Scheduler	0.31

Issue Bound Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 3.2 cycles. This might leave hardware resources unutilized and lowering overall performance, for more details refer to the Warp Scheduler section. It could be worthwhile to increase the number of eligible warps, for example by increasing occupancy or by ensuring enough independent work is ready to be issued. Ideally one eligible warp per cycle, Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp, no instruction can be issued. Each scheduler can issue one instruction every 1.00 cycles, but this workload only issues an instruction every 3.20 cycles. This might be caused by a lack of eligible warps, avoid possible issue inhibitors such as highly different execution durations per warp. Reducing stalls indicated on the Warps Per Scheduler chart and Warp Stall Sampling section can help tune.

Active Warps Per Scheduler [warp]	11.60
Eligible Warps Per Scheduler [warp]	1.64

Warps Per Scheduler

Warp State Statistics

Analysis of the states in which all resign warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive issued instructions. A high number of active cycles may warp pipelines is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the scheduler fail to issue every cycle. When not able to issue from a warp, this chart shows the reasons why the warp is not eligible to issue its next instruction.

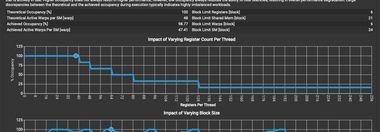
Warp Cycles Per Issued Instruction [cycle]	38.20	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	38.21	Avg. Not Predicated Off Threads Per Warp	31.02

Stc Thrashe Stalls On average, each warp of this workload spends 25.2 cycles being stalled waiting for the MIO (memory input/output) instruction queue to be not full. This stall reason is high in cases of frequently issuing instructions to either the MIO or TEX pipelines, or heavily issuing memory instructions that take more than two cycles to complete. Consider to issue fewer instructions, or to use fewer but wider loads can reduce pipeline pressure. This stall type represents about 65.9% of the total average of 38.2 cycles between issuing two instructions.

Section: Warp State Statistics

Warp Stall Check the Warps Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Kernel Profiling Guide provides more details on each stall reason.

Warp State (All Cycles)



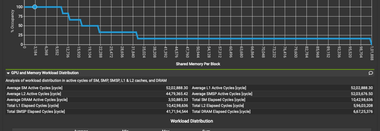
Instruction Statistics

Opcode Category Chart

Statistics of the executed low level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on specialized hardware units, which might limit overall performance if other functional units are idle. Instructions/Opcode and Executed Instructions are measured per thread, while Cycles per opcode and Cycles spent in system calls.

Executed Instructions [inst]	13,91,05,920	Avg. Executed Instructions Per Scheduler [inst]	16,13,824
Issued Instructions [inst]	13,91,14,816	Avg. Issued Instructions Per Scheduler [inst]	16,13,836.31
Local Memory Spilling Requests [inst]	0	Shared Memory Spilling Requests [inst]	0

Executed Instruction Categories



NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

NUMA information is not available on the target system.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing efficient launch configuration maximizes device utilization.

Block Size	4,096	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	104	Static Shared Memory Per Block [Kbyte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [Kbyte/block]	0
Threads [thread]	15,48,576	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	34.13	Shared Memory Configuration Size [Kbyte]	65.54
Uses Green Contexts	0	Grid Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all		

Occupancy

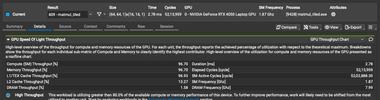
% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

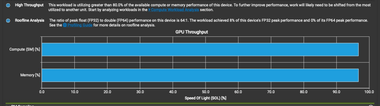
Theoretical Occupancy [%]	100	Block Limit Registers [block]	8
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	32
Achieved Occupancy [%]	96.70	Block Limit Warps [block]	32
Achieved Active Warps Per SM [warp]	47.41	Block Limit SM [block]	24

Impact of Varying Register Count Per Thread

Impact of Varying Block Size



Impact of Varying Shared Memory Usage Per Block



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMIT, SMSP, L1 & L2 caches, and DRAM.

Average SM Active Cycles [cycle]	5,232,888.50	Average L1 Active Cycles [cycle]	5,232,888.50	Average L2 Active Cycles [cycle]	5,232,888.50
Average L2 Active Cycles [cycle]	4,679,383.42	Average SMSP Active Cycles [cycle]	5,232,678.18	Average SMIT Active Cycles [cycle]	5,232,678.18
Average SMIM Active Cycles [cycle]	5,0,680.13	Total SM Elapsed Cycles [cycle]	1,04,58,577.00	Total L1 Elapsed Cycles [cycle]	1,04,58,577.00
Total L2 Elapsed Cycles [cycle]	9,36,77,154.00	Total L1 Elapsed Cycles [cycle]	5,94,23,328	Total SMSP Elapsed Cycles [cycle]	4,17,15,276

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	5,232,888.50	5,134,440	5,311,865	10,465,777
SMSP Active Cycles	5,232,678.18	5,134,664	5,311,139	10,464,139
L1 Active Cycles	5,232,888.50	5,134,440	5,311,865	10,465,777
L2 Active Cycles	4,679,383.42	4,618,025	4,676,693	9,357,767
Global Cycles Cycles	5,26,891.33	5,20,720	5,31,573	10,53,783

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the scheduler fails to issue every cycle.

Branch Instructions [inst]	21,95,488	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.2	Avg. Divergent Branches [branches]	0

Warp Stall Sampling (Not-issued Samples)

Location	Value [%]
matmul_tiled.cu:42 (0x7f0e0)	12,906
matmul_tiled.cu:42 (0x7f0a0)	5,965
matmul_tiled.cu:42 (0x7f0c0)	4,496
matmul_tiled.cu:42 (0x7f0b0)	4,215
matmul_tiled.cu:42 (0x7f0d0)	3,713

Most Instructions Executed

Location	Value [%]
matmul_tiled.cu:42 (0x7f0a0)	20,97,152
matmul_tiled.cu:42 (0x7f0b0)	20,97,152
matmul_tiled.cu:42 (0x7f0c0)	20,97,152
matmul_tiled.cu:42 (0x7f0d0)	20,97,152
matmul_tiled.cu:42 (0x7f0e0)	20,97,152

Follow the rules outputs to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual rules in the rules section to focus on selected performance aspects and make profiling faster.